CSE 211 – Data Structures

Term Project

PROJ-18: Airline Crew Rostering

Yeditepe University – Spring 2026

1 Problem Statement

Design and implement an airline crew rostering system that assigns flight crews to flights while respecting rest periods, qualification requirements, and working hour limits. The system must build pairings — multi-day trip sequences that start and end at a crew member's base — and then match crew members to pairings so that all flights are covered. The solution must comply with aviation regulations, handle fairness through seniority-based bidding, and support disruption recovery for delayed or canceled flights.

2 Core Concepts

- **Pairing:** A sequence of flights forming a round trip that starts and ends at a crew base, typically spanning 1 to 4 days. Each pairing is a self-contained work assignment.
- **Duty Period:** A block of consecutive working time within a pairing, limited to 10–14 hours depending on regulations. A pairing may contain multiple duty periods separated by rest.
- **Rest Requirement:** A minimum number of hours off between consecutive duty periods, ensuring crew members have adequate recovery time.
- **Qualification:** Crew members must be certified for the specific aircraft type they operate. A crew member without the required qualification cannot be assigned to that flight.
- **Deadheading:** A crew member traveling as a passenger on a flight to reposition to a different airport for their next duty assignment. Deadheading counts as cost but not duty time.

3 Key Challenges

- **Pairing Generation:** Enumerating legal flight sequences that form valid pairings is combinatorially explosive and requires pruning infeasible branches early.
- **Crew-Pairing Matching:** Assigning pairings to crew members such that every flight is covered exactly once while respecting qualifications and base assignments.
- **Cumulative Limits:** Tracking and enforcing weekly and monthly flight hour limits for each crew member across multiple pairings.

- **Fairness:** Implementing seniority-based bidding where senior crew members have priority in selecting preferred pairings and schedules.
- **Disruption Handling:** Reassigning crew when flights are delayed or canceled, minimizing cascading disruptions while maintaining regulatory compliance.

4 Sample Input

```json
{
  "flights": [
    {"id": "FL100", "from": "IST", "to": "ESB", "depart": "2026-04-01T08:00", "arrive": "2026-04-01T09:15", "aircraft": "A320"},
    {"id": "FL101", "from": "ESB", "to": "ADB", "depart": "2026-04-01T10:30", "arrive": "2026-04-01T11:45", "aircraft": "A320"},
    {"id": "FL102", "from": "ADB", "to": "IST", "depart": "2026-04-01T14:00", "arrive": "2026-04-01T15:20", "aircraft": "B737"},
    {"id": "FL103", "from": "IST", "to": "ADB", "depart": "2026-04-02T07:00", "arrive": "2026-04-02T08:20", "aircraft": "A320"},
    {"id": "FL104", "from": "ADB", "to": "ESB", "depart": "2026-04-02T10:00", "arrive": "2026-04-02T11:30", "aircraft": "B737"}
  ],
  "crew": [
    {"id": "CR01", "role": "captain", "base": "IST", "qualifications": ["A320", "B737"], "seniority": 15},
    {"id": "CR02", "role": "captain", "base": "IST", "qualifications": ["A320"], "seniority": 8},
    {"id": "CR03", "role": "first_officer", "base": "ESB", "qualifications": ["A320", "B737"], "seniority": 5},
    {"id": "CR04", "role": "first_officer", "base": "IST", "qualifications": ["B737"], "seniority": 3}
  ],
  "regulations": {
    "max_duty_hours": 12,
    "min_rest_hours": 10,
    "max_flight_hours_monthly": 100
  }
}
```

Listing 1: data/input_sample.json

For full project requirements, STL restrictions, submission format, and grading criteria, refer to ***base.pdf*** *(available on the Releases page).*